Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*High-Precision QCM-Based Real-Time Liquid Density Sensing System Utilizing functional coating material and Machine Learning-Enhanced Impedance Analysis*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (x), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics			
	Medium Category	Small Category	Medium Category	Small Category
	Measuring Instruments	*Measurement Sensors and Components*		
Concept of Proposed Technology	*The technology utilizes a Quartz Crystal Microbalance (QCM) surface-modified with functional films. Unlike traditional QCMs that struggle to separate density and viscosity, this system employs a differential approach with two sensors of varying film thicknesses to isolate the liquid trapped in pores, which behaves as a rigid mass. This is combined with Machine Learning (ML) regressors to analyze multi-parameter impedance data (resistance, reactance, phase), enabling simultaneous and highly accurate determination of fluid properties even in complex mixtures.*			
Rationale for Joint R&D Project	*Achieving industrial-grade reliability requires a synergy between Fraunhofer's expertise in advanced material synthesis and physical modeling, and a Korean SME's capabilities in system integration, portable electronics, and microfluidics. Collaboration is essential to transition from laboratory-scale verification to a field-ready, cost-effective monitoring solution for industries like semiconductor manufacturing and chemical processing*			
Key Words	1. *Quartz Crystal Microbalance (QCM)*			

	2. Liquid Density Measurement 3. Differential Measurement 4. Machine Learning Readout 5.

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	*Development of stable functional coatings with controlled porosity (50-70%) and optimized anchoring to prevent adsorption interference. Implementation of a Machine Learning-based inversion algorithm to decouple density and viscosity from impedance spectra. Integration of the sensor into a portable, temperature-controlled hardware platform for real-time monitorin*			
R&D Target Specifications	*Density Measurement Range: Up to 2.18 g/cm³. Accuracy: Average error of less than ±0.005 g/cm³ using the differential approach Frequency Stability: Repeatability within ±5 Hz. Thermal Stability: Integrated control maintaining 25 ± 0.3°C*			
R&D Plan	*Phase I (Pre-study, 6 months): Focus on the feasibility of functional film deposition on 5-MHz crystals and initial validation of the differential frequency shift model in controlled lab environments. Phase II (Main R&D, 30 months):- Months 7-18: Optimization of film ancoring for chemical resistance and development of the ML regressor using impedance parameters. - Months 19-30: Prototype fabrication including oscillator circuits and flow cells, followed by industrial validation in relevant environments to reach pilot-scale maturity*			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Fraunhofer Institute: Leading the design and fabrication of QCM sensors, performing high-resolution physical characterization, and developing the AI/ML models for multivariate impedance analysis.*
Expected R&R of Korean SME Partner	*Korean SME: Responsible for developing the portable hardware (Hartley oscillators, digital counters), designing the microfluidic measurement cells, and managing the industrial testing and market-specific validation within the Korean manufacturing sector.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The project will deliver a digitized, high-resolution density sensing platform that is significantly more compact and cost-effective than traditional laboratory densitometers. By overcoming the historical "viscosity-density entanglement" of QCM sensors through material innovation and ML, the system enables Industry 4.0 real-time process control. This technology will provide Korean SMEs with a competitive edge in providing high-performance monitoring tools for the global density sensor and chemical markets*

6. References

[1] D. Liu und L. Cao, „Simultaneous Measurement System of Liquid Density and Viscosity Using Steel Tuning Fork“, IEEE Sensors Journal, Bd. 23, Nr. 6, S. 5577–5584, März 2023, doi: 10.1109/JSEN.2023.3243236.

[2] J. Schwarzbeck, R. Neuder, M. Späth, und A. Jiménez-Sáez, „Scalable mm-Wave Liquid Crystal Reconfigurable Intelligent Surfaces based on the Delay Line Architecture“, 16. Januar 2026, arXiv: arXiv:2601.11307. doi: 10.48550/arXiv.2601.11307.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization